Bayu Febriansyah

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Education

University of Illinois Urbana-Champaign — Expected May 2027
M.S. in Aerospace Engineering; Current GPA: 3.83/4.00 — *Urbana, Illinois*

Bandung Institute of Technology (ITB) — Apr 2023
B.S. in Aerospace Engineering; Cum Laude; GPA: 3.64/4.00 — *Bandung, Indonesia*

Relevant Experience

3D Printing Engineer — May 2026 – Present
Illinois MakerLab — *Champaign, Illinois*
- Designed and manufactured mechanical components from client requirements and CAD models; designed fixtures, assembled and calibrated robots, and coordinated production and shipping of 200+ printed-part sets for IEEE lab kits.

Undergraduate Researcher — Oct 2022 – Apr 2023
Lightweight Structures Laboratory — *Bandung, Indonesia*
- Developed and benchmarked nonlinear finite element models in LS-DYNA; used parametric and sensitivity analyses to characterize axial collapse mechanisms, energy absorption, and design tradeoffs in lightweight, thin-walled structures.

Flight Data Engineering Intern — Jul 2022 – Sep 2022
PT Aering — *Bandung, Indonesia*
- Designed and implemented a Python flight-reconstruction workflow, integrating coordinate correction, altitude and heading processing, and map-based checks to generate X-Plane simulation input for takeoff and landing visualization.

Mechanical Engineering Intern — Jun 2021 – Aug 2021
Gexcon — *Jakarta, Indonesia*
- Conducted ANSYS CFD simulations to characterize fire behavior and propagation under industrial plant accident scenarios; interpreted results to support hazard evaluation, plant risk assessment, and safety engineering consulting.

Selected Projects

Tape Spring Hinge | *Space Structures Laboratory, UIUC* — 2026
- Compared Abaqus modal and nonlinear models with DIC geometry, MATLAB FFT vibration analysis, and three-point bend tests; assessed agreement across three vibration modes and documented the nonlinear model's convergence limit.

Carbon-Epoxy Composite I-Beam | *Baur Research Group, UIUC* — 2026
- Led fabrication from the team's numerical design, adapting layup, adhesive-film bonding, tooling, and cure restraint to available equipment; produced a Nomex-core beam for testing and an aluminum-honeycomb beam in a later build.

Ultrathin Bistable Composite Boom | *Space Structures Laboratory, UIUC* — 2026
- Built MATLAB laminate stiffness and strain-energy models; screened 5,476 cases, identifying transition energy and coiled geometry tradeoffs. Fabricated a baseline boom and compared modeled stable shapes with stereo DIC curvature.

Impact Analysis of Functionally Graded Crash Boxes | *Undergraduate Thesis, ITB* — 2023
- Compared five taper ratios under axial impact. Identified a graded multicell design with simulated specific energy absorption (per crushed mass) 124% above a conventional square tube and 11.8% above a conventional multicell tube.

Technical Skills

Programming and Data Processing: MATLAB, Python (NumPy, pandas, SciPy), Excel (VBA, Power Query).
CAD and Design: Siemens NX/NX CAM, SOLIDWORKS, Fusion 360/Fusion Manufacture, GD&T.
Engineering Simulation and Analysis: LS-DYNA, Abaqus, ANSYS Mechanical/Fluent/CFX, Patran/Nastran.
Languages: Indonesian (native) and English (fluent, TOEFL iBT 109/120, IELTS 8.0/9.0).

Honors, Awards, and Activities

Tau Beta Pi Engineering Honor Society (2026): Member; recognizes academic excellence and exemplary character.

Indonesia Endowment Fund for Education (LPDP) (2025): Awardee; Indonesian government graduate scholarship.

Dean's List, Bandung Institute of Technology (2023): Awardee; ranked among the top 5% of students consistently.

Memberships: Society for the Advancement of Material and Process Engineering (SAMPE), UIUC Chapter; American Institute of Aeronautics and Astronautics (AIAA), UIUC Chapter; Aerospace Graduate Student Advisory Committee (AeroGSAC); Indonesian Students Association (PERMIAS), UIUC; Indonesian Aerospace Engineering Alumni Association (IAP ITB).